

▲ AEGIS

Alert Enrichment, Grouping & Intelligence Scoring

Threat-intelligence correlation and alert prioritisation, with an evidence trail IBM Bob can be asked about.

Team Blue Vector · Track: AI

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The problem: four consoles, four severity scales, one analyst

Volume

Thousands of alerts a day across SIEM, network sensors, geospatial feeds and written intelligence. Nobody reads them all, so triage is "whatever sorts to the top".

Incomparable severity

The SIEM says critical, the sensor says err, the geo feed says P2, the report says HIGH. None knows the asset or what else is happening.

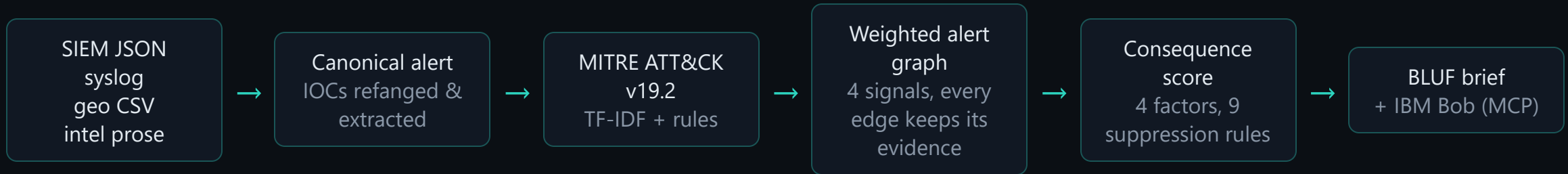
Asymmetric cost

A false positive costs an analyst-hour. A missed intrusion costs the mission. Real intrusions arrive as three unremarkable alerts in three consoles.

[illegible]

What the analyst actually starts the shift with.

AEGIS: triage as a graph problem with an evidence trail



Four correlation signals per alert pair

- **Shared indicator** ≤ 0.40 — rarity-weighted (IDF over the corpus); a shared SHA-256 beats a shared DNS server
- **Temporal proximity** ≤ 0.20 — exponential decay, no hard window
- **Asset adjacency** ≤ 0.15 — same host, user, subnet, or one alert naming the other's asset
- **Tactic progression** ≤ 0.25 — *directional* kill-chain ordering: Initial Access → Execution scores, the reverse does not

Why it is not "group by source IP"

- Incidents are connected components above a threshold — count emerges from evidence, not a chosen k
- Ordering only chains alerts already tied by an entity; timing alone is coincidence
- Priority is consequence-driven: a medium-confidence hit on a domain controller outranks a certain one on a print server
- Vendor severity is preserved, shown, and never scored

Architecture: one state, two surfaces

Pipeline (Python 3.10, ~3 s for 521 alerts)

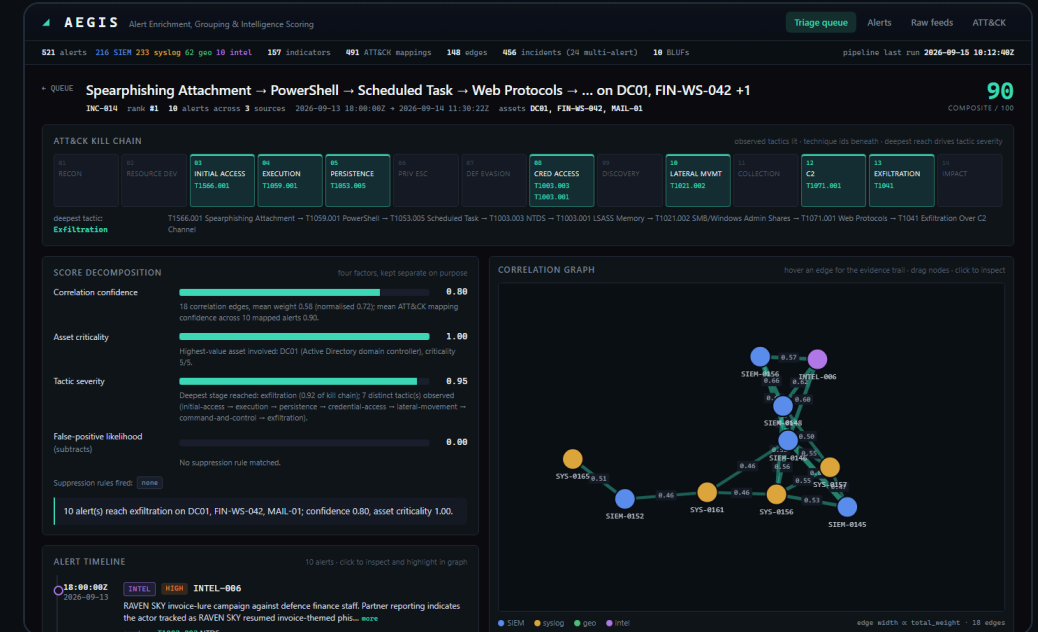
Ingest adapters → SQLite (stdlib) → ATT&CK mapper (numpy TF-IDF over the pinned STIX corpus, 697 techniques) → correlation engine (NetworkX) → prioritiser → BLUF generator (template; watsonx.ai Granite refinement validated against evidence)

Service layer → FastAPI + MCP server

The console and IBM Bob read the same SQLite through the same functions. Neither computes anything, so they cannot disagree.

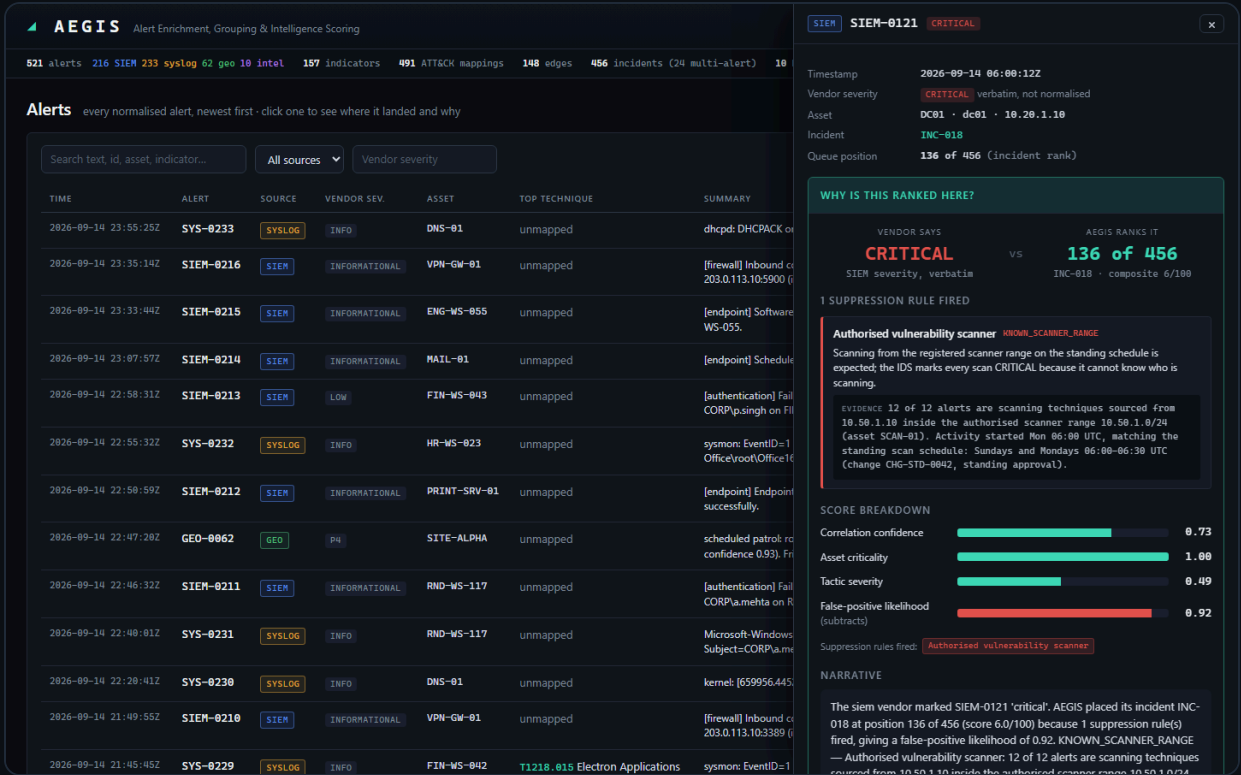
Console: plain HTML/JS, no build step

Force-directed correlation graph on canvas, evidence on edge hover, kill-chain strip, score decomposition, BLUF panel, "why is this ranked here?"



INC-014: phishing → PowerShell → C2 → DC01 credential theft → exfiltration. 10 alerts, 3 sources, 18 evidential edges.

The closer: "The SIEM marked this CRITICAL. Why is it ranked 136th?"



What AEGIS answers

Position 136 of 456. Rule KNOWN_SCANNER_RANGE: 12 of 12 alerts are scanning techniques from 10.50.1.10 inside the authorised scanner range, Monday 06:00 UTC, matching the standing schedule. Correlation confidence 0.73, asset criticality 1.00 — and a false-positive likelihood of 0.92.

What would change its mind

A different source address, a different account, activity outside the window, or any technique beyond scanning. The rule is narrow on purpose.

The same answer is what IBM Bob gives via why_deprioritised. It exists only because every edge and every score kept its evidence.

IBM Bob is the analyst's interface, not a footnote

Six read-only MCP tools (Python MCP SDK, stdio)

TOOL	THE ANALYST ASKS
get_priority_queue	"What should I look at first this shift?"
explain_correlation	"Why are SIEM-0146 and SYS-0161 the same incident?"
get_bluf	"Give me the BLUF for INC-017."
why_deprioritised	"Why is critical alert SIEM-0121 ranked so low?"
search_by_technique	"Show me everything mapped to T1059."
get_incident	"Walk me through INC-014."

Built with Bob

Bob was used throughout the build: planning the module contracts, implementing the adapters and correlation signals, reviewing the tactic matrix, and driving the terminal for the pipeline runs.

watsonx.ai Granite (optional)

Rewrites the BLUF narrative from the structured evidence under a strict JSON schema. Hallucinated alert IDs are rejected; the template path is what the demo runs on.

Remove Bob and...

the analyst loses their primary conversational interface into the evidence trail — not a summary paragraph.

Measured against ground truth, not asserted

521 synthetic alerts · 8 planted scenarios (6 intrusions, 2 look-alike benign) · 466 noise · `python -m src.eval.evaluate`

SCENARIO	TRUTH	COVERAGE	PURITY	QUEUE POSITION	SUPPRESSION
Invoice phishing → PowerShell → C2 → DC01 NTDS → exfiltration	intrusion	1.00	1.00	1	—
UAS + RF emitter → rogue AP → OT VLAN → PLC write (SITE-ALPHA)	intrusion	0.86	1.00	2	—
VPN password spray → account takeover → AD/share enumeration	intrusion	1.00	1.00	3	—
CVE-2023-22518 → web shell → cron → reverse shell	intrusion	1.00	1.00	4	—
Shadow-copy deletion → masquerading → mass encryption	intrusion	1.00	1.00	5	—
Insider staging → personal cloud upload (ambiguous)	intrusion	1.00	1.00	6	—
Sysadmin rollout inside change window	benign	0.75	1.00	117	MAINTENANCE_WINDOW ADMIN_FROM_ADMIN_WORKSTATION
Authorised scan, 8 vendor-CRITICAL alerts	benign	1.00	1.00	136	KNOWN_SCANNER_RANGE

6 / 6

intrusions in the top 6 of 456

2 / 2

benign look-alikes below every intrusion

0

noise alerts in any planted incident

55 / 55

planted alerts mapped to ATT&CK

Results & impact

7 / 8

Harvest misses a HUMINT report with no technical indicator and one geo track linked only by time and place — the cost of refusing to chain on timing alone. Carnus is

Team Blue Vector — and where this goes

MEMBER	BUILT
Het Khatusuriya (lead)	Corpus & ground truth, IOC extraction, tactic ordering matrix, suppression rules, asset inventory
Manan Modi	ATT&CK mapping, correlation graph & signal weights, BLUF generation, evaluation harness
Mohammad Safik	Analyst console: queue, correlation graph, kill-chain view, BLUF panel, screenshots
Samarth Bhalala	Canonical schema, four ingest adapters, SQLite store, FastAPI, MCP server for Bob

Beyond the hackathon

- Streaming ingestion with incremental graph updates
- Suppression knowledge from the CMDB and change system, not code
- Analyst feedback loop to recalibrate signal weights per enclave
- Bob as the first-line triage assistant on the watch floor

`github.com/het-khatusuriya/bob-ai-hackathon-blue-vector`

`python scripts/demo.py → http://localhost:8000`